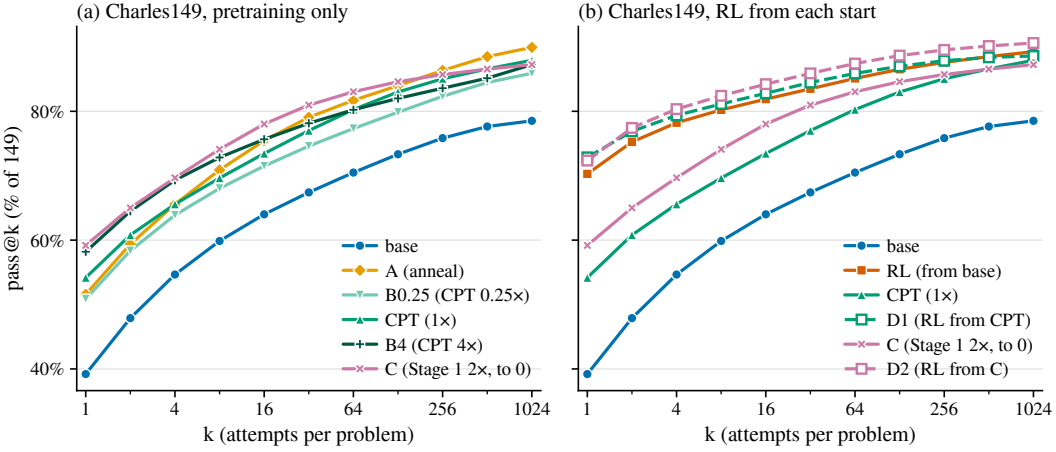
(a) Charles149, pretraining only
(b) Charles149, RL from each start
pass@k (% of 149)
80%
60%
40%
1
4
16
64
256
1024
k (attempts per problem)
base
A (anneal)
B0.25 (CPT 0.25×)
CPT (1×)
B4 (CPT 4×)
C (Stage 1 2×, to 0)
RL (from base)
D1 (RL from CPT)
C (Stage 1 2×, to 0)
D2 (RL from C)